

# **Terahertz photocurrent probe of quantum geometry and interactions in magic-angle twisted bilayer graphene**

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## Table of Contents:

|                                                                                                                                                                        |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>S1 - Comparison with the photo thermoelectric effect .....</b>                                                                                                      | <b>2</b>  |
| <b>S2 – Polarization Dependence in D1.12 and D1.03 samples .....</b>                                                                                                   | <b>2</b>  |
| <b>S3 – Wavelength dependence of photoresponse in D1.12 .....</b>                                                                                                      | <b>3</b>  |
| <b>S4 – Additional checks of the polarization dependence .....</b>                                                                                                     | <b>3</b>  |
| <b>S5 – Polarization dependence in non-centrosymmetric 2D materials, near-field enhancements, and junction contributions .....</b>                                     | <b>4</b>  |
| <b>S5.a - Bilayer graphene – Inhomogeneous devices and photoactive junctions far from gold contacts .....</b>                                                          | <b>4</b>  |
| <b>S5.b - Single Layer Graphene – High quality device with photoactive junctions close to near-field enhancements .....</b>                                            | <b>5</b>  |
| <b>S5.c - D1.02 Magic-angle twisted bilayer graphene – contributions to the photoresponse by near-field enhancements and isolating the second-order response .....</b> | <b>5</b>  |
| <b>S6 – Theoretical analysis of and shift current calculations .....</b>                                                                                               | <b>6</b>  |
| <b>S6a - Continuum model for twisted bilayer .....</b>                                                                                                                 | <b>6</b>  |
| <b>S6b - Relating experimental and theoretical quantities .....</b>                                                                                                    | <b>7</b>  |
| <b>S6c - Symmetry dependence .....</b>                                                                                                                                 | <b>8</b>  |
| <b>S6d - Filling dependence .....</b>                                                                                                                                  | <b>9</b>  |
| <b>S6e - The Role of interactions .....</b>                                                                                                                            | <b>10</b> |
| <b>S6f - The Role of strain .....</b>                                                                                                                                  | <b>10</b> |
| <b>S7 – Quantum transport measurement of the <math>2\omega</math> response in twisted bilayer graphene .....</b>                                                       | <b>11</b> |
| <b>S8 – Temperature dependent quantum transport measurements characterizing the gaps at full filling .....</b>                                                         | <b>11</b> |
| <b>S9 – Alignment angles of twisted bilayer graphene to hBN extracted from optical images .....</b>                                                                    | <b>11</b> |
| <b>S10 – Tight-binding calculations of our twisted bilayer graphene encapsulated with hexagonal-boron nitride devices .....</b>                                        | <b>12</b> |
| <b>S11 – Second-order response in D 1.5 and measurement of C3 symmetry .....</b>                                                                                       | <b>13</b> |
| <b>S12 – Measurements between different contact configurations in magic-angle samples .....</b>                                                                        | <b>14</b> |
| <b>S13 – Terahertz photodetection in magic-angle twisted bilayer graphene .....</b>                                                                                    | <b>15</b> |
| <b>Supplementary Figures .....</b>                                                                                                                                     | <b>16</b> |

## S1 - Comparison with the photo thermoelectric effect

In graphene-based heterostructures, the long lifetime and cooling time of photoexcited hot electrons ensure that its photoresponse exhibits a strong photo thermoelectric (PTE) effect which is driven by the Seebeck coefficients of the materials and temperature gradients induced by heating of the electron gas via photoexcitation<sup>1,2</sup>. Given the large Seebeck coefficients predicted in magic-angle twisted bilayer graphene<sup>3</sup> and a dominant contribution to the terahertz response observed in graphene devices<sup>4,5</sup>, one may expect a manifestation of the PTE effect. We explore this possibility by comparing the photoresponse measured in our devices with the functional form expected for the Seebeck coefficient using the semi-classical Mott formula<sup>6</sup> (S1)

$$S(E_F) = \frac{\pi^2 k_B T}{3e} \frac{1}{G} \frac{dG}{dV_g} \frac{dV_g}{dE_F}, \quad (S1)$$

where  $G$  is the 2-probe conductance measured between the same pair of contacts used in photovoltage measurements, and  $V_g$  is the gate voltage. Using the density of states (DOS) obtained from the calculated band structures (Fig. 1a main text inset), we obtain the filling-dependent Seebeck coefficient  $S(\nu)$ . It is plotted in Fig. S2b for different temperatures calculated from the 2-probe resistance in D0.94, using the same pair of contacts as photoresponse measurements. In general, the leading contribution to the Mott formula is the transconductance term which is calculated from the gate dependence of the resistivity and results in the oscillatory structure imposed by the integer filling features observable in quantum transport. Figure S2a plots the photoresponse measured in D0.94; we plot the polarization-dependent component of the response  $V_{\text{Lin}_L1}$  obtained by taking  $V_{\text{Lin}_L1} = V_{\text{PV}} - V_{\text{P}} = 135 - V_{\text{P}}$ . One can see that the two responses are rather different. The photoresponse exhibits a complex form, peaked at zero doping and increasing gradually approaching full filling  $\nu = 4$ . In the PTE contribution, which would follow the estimated Seebeck coefficient, strong oscillations occur likely originating from the changes in transconductance close to integer fillings. This comparison along with the complex polarization dependence (Fig. 2 of main text) provides strong evidence for a contrasting mechanism to the PTE effect.

## S2 – Polarization Dependence in D1.12 and D1.03 samples

Figure S3 plots the full polarization and filling factor ( $\nu$ ) dependence for two other magic-angle samples presented in the main text, D1.12 and D1.03. The full polarization dependence presented in Fig. 1g,h is plotted in Fig. S3a,d. In both samples the unique behaviour at CNP is observed, that is a peak response at CNP for one polarization direction that evolves smoothly from a sign-changing response. In addition, we find complex polarization dependence can be observed at higher doping levels, especially close to full filling. Like D0.94 (Fig. 2a), the polarization phase ( $\alpha$ ) changes when tuning the filling factor. This is demonstrated in the colour maps which plot the polarization-dependent component  $V_{\text{Lin}}$  and the white lines that track the changes in  $\alpha$ , as shown in Fig. 2 of the main text. The raw data are plotted in Fig. S3c/f tracking the changes in  $\alpha$  as the filling is changed. Although both dependencies are rather different, they demonstrate a phase drift through CNP (shown for D1.03 in Fig. S3). Some similarities between the two include negative pockets of resistance between integer fillings appearing with polarization and sharp slips in  $\alpha$  close to  $\nu = 4$ . In summary, Fig. S3 demonstrates that complex polarization-dependent photocurrents manifest profoundly at THz frequencies in magic-angle twisted bilayer graphene.

### S3 – Wavelength dependence of photoresponse in D1.12

As mentioned in the main text, most measurements presented in D1.12 are presented for 0.7 THz (2.7 meV) excitation wavelength rather than 2.5 THz (10 meV). This is because, 0.7 THz showed a much larger response (Fig. 1 of main text) and clearer signatures of the second-order photocurrents including the complex polarization dependence close to the charge neutrality (Fig. 1a of the main text) and drifts in the polarization phase (Fig. S3). The differences are shown in Fig. S4a which compares the photovoltage measured with the same pair of contacts for two different wavelengths 2.5 and 0.69 THz for a fixed polarization direction. Not only is the photoresponse an order of magnitude larger at 0.69 THz, but the dependence on the filling factor is different. The peaked response at CNP is significantly more pronounced and additional features can be discerned, including oscillatory structures at integer fillings and negative/positive pockets of photoresponse between  $\nu = 3-4$ . The difference in polarization dependence can be seen by comparing the colour maps in Fig. S4b and Fig. S3b, which plots the polarization-dependent component of the photoresponse  $V_{\text{Lin}}$  for 2.5 THz respectively. Indeed, there is clearly some polarization dependence, but the polarization phase  $\alpha$  is significantly more rigid, varying little compared to the case of 0.69 THz. The differences can be seen clearer in Fig. S4c which compares the polarization phase for the two wavelengths and same pair of contacts. In 0.69 THz, the polarization drifts smoothly over  $90^\circ$  from  $\nu = 0-4$  whilst in 2.5 THz it is nearly constant with some minor variations. Furthermore, the peak-like structures and drifts at integer filling are clearly only discerned at 0.69 THz (Fig. S4c). The complex wavelength dependence provides further evidence for the origin in second-order response which is very sensitive to wavelength<sup>7,8,9</sup>. Moreover, the significantly larger response suggests we approach the intrinsic THz resonances of the flat bands in magic-angle TBG (Fig. S16) and begin to dominate over any extrinsic polarization axes that may be overwhelming the response at 2.5 THz. Finally, we note that the differences in wavelength may also be due to competing mechanisms, for example, the rigid behaviour of the polarization phase mimics something expected from an antenna (See Supplementary Section 5), while moving to shorter wavelengths we approach the intrinsic bulk response of the system with two principal axes.

### S4 – Additional checks of the polarization dependence

Polarization-resolved measurements can be challenging with many artefacts arising. One of the artefacts is that of beam precession, where the photoresponse oscillates or changes owing to a change in spatial position induced by rotating optical elements (waveplates/polarizers) rather than originating from a change in polarization itself. In general, such effects should be minor in our experiment because the focused THz beam is much larger than our device size. Nonetheless, we performed additional checks to verify that the beam precession is minimal and does not significantly influence the measured response. For this purpose, we performed a full  $360^\circ$  degree polarization dependence of the photocurrent using our polarizer. If our measurement is a true polarization effect, then the curves should be identical for polarization directions separated by  $180^\circ$ . Figure S5a-c plots photocurrent measurements performed for different polarization directions and their  $180^\circ$  rotations in D1.12 at 0.7 THz. The solid and dotted lines for each polarization direction represent data with the same polarization direction, but with the polarizer rotated by  $180^\circ$ . Notably, in all cases, the polarization dependences exhibits almost identical photoresponses, except for minute changes. These results demonstrate that our measurements track a change in polarization state whilst beam precession is at a minimum, and justifies the analysis of the polarization dependence within a limited range of  $\theta_p$  presented for specific measurements in the manuscript.

## **S5 – Polarization dependence in non-centrosymmetric 2D materials, near-field enhancements, and junction contributions**

A polarization-dependent photoresponse is typically observed in many different types of mesoscopic devices even in systems that pertain inversion symmetry. Such effects usually originate from extrinsic symmetry breaking of the device architecture such as device edges<sup>10</sup>, or near-field enhancements at metallic leads<sup>11</sup>. In asymmetric structures, polarization-dependent photocurrents can be induced that even changes sign with polarization direction<sup>12,13</sup>. Their origin lies in the near-field enhancement of optical fields at the edges of gold contacts. Such near-field enhancements are also polarization-sensitive via the so-called lightning rod effect and can produce a polarization dependent response. To understand possible contributions that could enter our measurements of TBG samples, we measured two control devices. The first, consisted of a single layer graphene (SLG) Hall bar device, with a mesa-like structure similar to that of our TBG devices. The second, is a bilayer graphene (BLG) device that is rather inhomogeneous owing to bubbles and contamination within the heterostructure and contains multiple PN junctions. The motivation for studying such samples is to understand whether inhomogeneity and/or asymmetric device mesas could cause behaviour mimicking the bulk photovoltaic effect.

### **S5.a - Bilayer graphene – Inhomogeneous devices and photoactive junctions far from gold contacts**

First, we present data measured on the BLG device. An optical image is presented in Fig. S6a. It consists of a graphite/hBN/bilayer graphene/hBN/graphite heterostructure. The two graphite contacts are marked in red and dark blue respectively and are used as top and bottom gates to gate the bilayer graphene channel. We note that heterostructure is rather inhomogeneous, with many bubbles in the active area and multiple PN junctions formed at the edge of graphite gates and close to gold contacts. The goal is to understand whether devices with many junctions could cause a non-trivial response. Figure S6c shows the photocurrent as a function of the bottom gate voltage  $V_{\text{Graphite}}$  for two different polarization directions that are orthogonal to each other. In general, both data have a similar gate dependence, featuring peaks close to zero doping and a function that resembles the Seebeck coefficient in graphene heterostructures, likely originating from the PTE effect<sup>14,2</sup>. The effect of polarization seems to cause only a constant offset in the measured photoresponse. By fixing the doping and varying the polarization angle (Fig. S6b inset) we observe a clear sinusoidal dependence indicative of a polarization dependent response in the system. However, measuring for different doping levels we find the axis of the polarization dependence is fixed. This is clearer seen in Fig. S6e which plots the polarization dependent component of the photocurrent ( $V_{\text{Lin}}$ ) as a function of gate voltage and polarization angle, similar to the maps presented in Fig. 2b, Fig. 5d and & Fig. S3b,e. It shows that the polarization phase  $\alpha$  remains fixed for all the doping levels contrasting greatly with data measured in TBG. Figure 2f plots the extracted coefficients from which we find L1 and L2 are rather different from the offset, D, and are almost entirely independent on the carrier doping. We interpret this behaviour in the following way. In our BG device, there are four junctions. Two of these are formed by the edge of graphite gates (J1 in Fig. S6a), and the other two by gold-BG junctions (J2 in Fig. S6a). Since graphite gates only tune J1, the gate-tuneable photoresponse we measure is likely dominated by those regions. Since the photo-thermoelectric effect is polarization insensitive, it makes sense that the polarization dependent components L1, L2 do not follow its gate dependence. On the other hand, the other two Au-BG junctions are not gate-tuneable. They can however exhibit some polarization dependence via the lightning rod effect, where near fields in the Au are enhanced for certain polarization direction, enhancing the absorption in graphene, and resulting in the coefficients L1, L2 that are independent on the graphite gate. This result indicates that the photo-thermoelectric effect – a mechanism known to dominate in graphene and graphene bilayers – is polarization

independent, and the only polarization dependence enters through near-field absorption enhancements close to gold contacts.

### **S5.b - Single Layer Graphene – High quality device with photoactive junctions close to near-field enhancements**

To further test our understanding of near-field enhancements, we studied a single layer graphene encapsulated with hexagonal-boron nitride device where the gold leads are close to graphite gates and can influence the photoresponse via near-field enhancements (Fig. S7a). The two-probe resistance measured as a function of gate voltage between contact pairs indicated in Fig. S7a is plotted in Fig. S7b revealing a single peaked response close to zero gate voltage when the Fermi-level is tuned to the Dirac point. Figure S7c plots the photovoltage ( $V_{PV}$ ) as a function of the gate voltage for three different polarization directions. For  $170^\circ$  the response is maximum, and its gate dependence mimics what is expected from the photothermoelectric effect changing sign at the Dirac point. Rotating the polarization response lowers the signal smoothly and exhibits a  $180^\circ$  periodicity expected for a polarization-dependent response (Inset of Fig. S7c). This behaviour mimics an antenna/lightning rod effect in-which the response is maximum for one direction and close to zero for the perpendicular when the near-field enhancements are suppressed. Indeed, the polarization dependence highlights this approaching zero for  $100^\circ$ . We note however that it also can change sign with the polarization (black curve in Fig. S7c). Nonetheless, tracking the polarization phase through plots of  $V_{LIN}(\theta, V_G)$  (Fig. S7d) shows that the polarization phase ( $\alpha$ ) is fixed for all the gate voltages confirming its origin in extrinsic absorption enhancements. Its origin in an antenna effect can be verified by fitting the polarization dependence completely with the function  $V_{PV} = L\sin^2\theta + D$  (red line in the inset of Fig. S7c). The  $\sin^2$  term captures the polarization dependence governed by an antenna, whilst the offset  $D$  captures a polarization independent response in the system (plotted in Fig. S7e). With this, the sign changing response with polarization can be understood as an interplay of an antenna enhanced junction with a polarization independent off-set of opposite sign. When the antenna response is small ( $\theta_p = 130^\circ$ ) the off-set ( $D$ ) starts to dominate the measurement resulting in a sign-change with polarization. Similar behaviour was observed in graphene PN junction devices previously<sup>41</sup>. In summary, polarization-dependent responses can be expected in any mesoscopic graphene device that preserves inversion symmetry. The key distinction with the second-order response reported in the main text is that the electron system can rotate the principal axes/polarization phase (Fig. 2d), whilst in these devices it is fixed for all the doping levels (Fig. S7d).

### **S5.c - D1.02 Magic-angle twisted bilayer graphene – contributions to the photoresponse by near-field enhancements and isolating the second-order response**

In this section, we present measurements and analysis of a fourth magic angle device D1.02 for which oscillations in the photoresponse could be observed baring some similarities with D1.03 (Fig. 5 of main text). The device is presented in Fig. S8a. In contrast to the other TBG devices, where the graphite gates are under the gold leads, an un-gated region of TBG exists at the gold-leads in D1.02. The junction, which typically always induces a photovoltage with a polarization dependence caused by near-field enhancements (as in the case of Fig. S6/7), is not gate-tuneable and manifests in our data as a polarization dependent background. This is seen clearly in Fig. S8b which plots the photovoltage measured for three directions. Similar to the case of Bilayer graphene (Fig. S6) the polarization offsets the data and follows a  $180^\circ$  periodicity characteristic of a polarization dependence. However, in contrast to the BLG case, we notice that the functional form with carrier doping is also polarization-sensitive. Therefore, we subtract the background to isolate the polarization dependent component of the photovoltage from the gold-un-gated region whose response is not representative of the second-

order photocurrents. Its influence on the polarization dependence can be seen in Fig. S8c which plots  $V_{\text{Lin}}(n, q_p)$  and Fig. S8e which plots the polarization phase as a function of filling factor  $a(n)$  accordingly. Indeed, despite the oscillatory structure in  $a$ , we notice that the polarization phase is rather rigid, oscillating weakly around a constant value  $= 80^\circ$  likely dominated by the near-field enhancements. This indicates that additional gate-tuneable contributions to the polarization phase exist, but are masked by the principal axes of near-field enhancements. According to our control measurements in BLG, this polarization-dependent background contributes to a gate-independent photovoltage offset (Fig. S6c). Therefore, we subtract linear fits to data of Fig. S8b (dashed lines) to isolate the gate-tuneable component of  $a$ . The corresponding  $V_{\text{Lin}}(v, \theta_p)$  maps are plotted in Fig. S8d and resemble now something closer to measurements performed in D1.03 which explicitly has global alignment between TBG and hBN. Specifically, sharp phase changes in  $a$  can be seen at various doping levels (Fig. S8f).  $V_{\text{PV}} - \text{Background Subtraction}$  is plotted for different filling factors in the vicinity of these drifts in Fig. S8g to visualize and verify the extraction of polarization phase (Fig. S8f). Although the interpretation of the data in D1.02 is slightly more convoluted than the other samples, these measurements and analysis highlight how multiple mechanisms may be contributing to the polarization phase in the THz regime of TBG. The seemingly less prominent second-order response in D1.02 may be due to its lack of strong global alignment to hBN and hence a weaker influence of the symmetry-breaking substrate essential for driving the second-order response.

## S6 – Theoretical analysis of and shift current calculations

In this section of the Supplemental Materials, we provide the details of the theoretical analysis described in the main text. Specifically, we compute the shift current in twisted bilayer graphene by following Ref. 7 and the references discussed therein. The analysis considers two twist angles,  $0.94^\circ$  and  $1.12^\circ$  where the band structure is calculated using the BM model<sup>15</sup>. Our analysis specifically considered flat-to-flat band transitions only, as flat-to-dispersive transitions are suppressed given the low experimental frequency. For a detailed discussion of flat-to-flat and flat-to-dispersive transitions, we refer the reader to Ref. 7.

### S6a - Continuum model for twisted bilayer

We consider the continuum model used in Ref.[7,16], which gives a Hamiltonian

$$H_0 = \sum_{\gamma=\{\zeta, \sigma\}} \int_{\Omega} d^2\mathbf{r} \psi_{\gamma}^{\dagger}(\mathbf{r}) \hat{H}^{(\zeta, \sigma)} \psi_{\gamma}(\mathbf{r}) \quad (\text{S1})$$

$$\hat{H}^{(\zeta, \sigma)} = \begin{pmatrix} H_{\zeta,1}(\mathbf{r}) & U_{\zeta}^{\dagger}(\mathbf{r}) \\ U_{\zeta}(\mathbf{r}) & H_{\zeta,2}(\mathbf{r}) \end{pmatrix} \quad (\text{S2})$$

where  $\Omega$  represents the moiré unit cell,  $H_{\zeta,l}$  are two by two matrices for valley  $\zeta$  in sublattice basis (A,B) and represents the intralayer Hamiltonian for layer denoted by  $l = 1, 2$ , and  $U_{\zeta}(\mathbf{r})$  represents the interlayer hopping matrix.

The intralayer part of the Hamiltonian,  $H_{\zeta,l}$  is given by the two-dimensional Dirac Hamiltonian expanded around the  $\mathbf{K}_{\zeta}^l$  point of the original graphene layer

$$H_{\zeta,l} = -\hbar v \left[ R \left( \pm \frac{\theta}{2} \right) (\mathbf{k} - \mathbf{K}_{\zeta}^l) \right] \cdot (\zeta \sigma_x, \sigma_y) + \Delta_l \sigma_z, \quad (\text{S3})$$

Where  $\mathbf{k}$  is a momentum in the Brillouin zone (BZ) of the original graphene layers,  $R \left( \pm \frac{\theta}{2} \right)$  is the  $2 \times 2$  two-dimensional matrix accounting for the rotation of layer  $l = 1(2)$  by an angle  $\pm \frac{\theta}{2}$  about the  $z$  axis with respect to initial AA stacked bilayer. We set  $\frac{\hbar v}{a} = 2.943$  (3.21) eV for  $\theta = 1.12^\circ$  ( $0.94^\circ$ ).

The last term,  $\Delta_l \sigma_z$  in above equation represents the sublattice energy offset for each layer which breaks inversion symmetry.

The moiré interlayer potential  $U_\zeta(\mathbf{r})$  can be approximated as

$$U_\zeta(\mathbf{r}) = \begin{pmatrix} u & u' \\ u' & u \end{pmatrix} + \begin{pmatrix} u & u' e^{-i2\pi\zeta/3} \\ u' e^{i2\pi\zeta/3} & u \end{pmatrix} e^{i\zeta \mathbf{G}_1^M \cdot \mathbf{r}} + \begin{pmatrix} u & u' e^{i2\pi\zeta/3} \\ u' e^{-i2\pi\zeta/3} & u \end{pmatrix} e^{i\zeta (\mathbf{G}_1^M + \mathbf{G}_2^M) \cdot \mathbf{r}} \quad (S4)$$

where  $u = 75$  (70 meV)  $u' = 115$  (85) meV for  $\theta = 1.12^\circ$  ( $0.94^\circ$ ) and  $\mathbf{G}_i^M$  are moiré reciprocal lattice vectors with  $\mathbf{G}_1 = \frac{2\pi}{a}(1, -\frac{1}{\sqrt{3}})$  and  $\mathbf{G}_2 = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}})$  being the reciprocal lattice vectors of original graphene layer.

We obtain energy dispersion and eigenstates by diagonalizing the Hamiltonian in  $\mathbf{k}$  space limited to mini BZ and consider the effect of interlayer term by coupling  $\mathbf{k}$  and  $\mathbf{k} + \mathbf{G}$  where  $\mathbf{G} = n_1 \mathbf{G}_1^M + n_2 \mathbf{G}_2^M$  for integers  $n_1, n_2$ .

### S6b - Relating experimental and theoretical quantities

We begin by detailing how microscopically computed shift current can be related to the experimentally relevant quantity photovoltage. Shift current is an example of a non-linear response of an electronic system to an external electric field. We can relate the applied electric field to the measured photocurrent  $j$  in general as

$$\begin{pmatrix} j_x \\ j_y \end{pmatrix} = \begin{pmatrix} \sigma_{xxx} & \sigma_{xxy} & \sigma_{xyx} & \sigma_{xyy} \\ \sigma_{yxx} & \sigma_{yyx} & \sigma_{yyx} & \sigma_{yyy} \end{pmatrix} \begin{pmatrix} E_x^2 \\ E_x E_y \\ E_y E_x \\ E_y^2 \end{pmatrix}, \quad (S5)$$

where we suppressed time and frequency dependence of the electric field. To relate it to the experiment it is helpful to express the polarization of electric field in the polar coordinate system of the crystallographic axis of the material, i.e.  $E_x = E \cos \phi$  and  $E_y = E \sin \phi$  with  $E$  denoting electric field magnitude. In such polar coordinates we have:

$$j_x = (\sigma_{xxx} \cos^2 \phi + (\sigma_{xxy} + \sigma_{xyx}) \cos \phi \sin \phi + \sigma_{xyy} \sin^2 \phi) E^2 \quad (S6)$$

$$j_y = (\sigma_{yxx} \cos^2 \phi + (\sigma_{yyx} + \sigma_{xyy}) \cos \phi \sin \phi + \sigma_{yyy} \sin^2 \phi) E^2 \quad (S7)$$

In the above Eqs.  $\sigma_{ijk}$  are the second-order conductivity tensor components, whose microscopic origins are explained in what follows. We note that the above relation applies to any non-linear second-order response – specific relation to shift current is only enforced when the coefficients  $\sigma_{ijk}$  are computed from microscopic principles.

In an experiment, the coordinate system is likely rotated with respect to the crystallographic axis of the system by an angle  $\beta$ . We thus can carry a standard similarity transform (rotation) of the crystallographic axis  $x, y$  to the experimental axis  $x', y'$ :

$$\begin{pmatrix} j'_x \\ j'_y \end{pmatrix} = \begin{pmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{pmatrix} \begin{pmatrix} j_x \\ j_y \end{pmatrix} \quad (S8)$$

Lastly, the experiment measures photovoltage, which we take as linearly proportional to the photocurrent, i.e.  $V_i \propto j_i$ . We use the terms photovoltage and photocurrent interchangeably in what follows. The misalignment of the crystallographic axis and the axis of the experiment is a free parameter, which we vary to optimize agreement with observations, but we keep it fixed for a given

device. Specifically, we choose the polarization direction ( $\theta_p$ ) for which the peaked response at CNP is observed to define the principle crystallographic axis  $\sigma_{xxx}$  corresponding to  $\beta = 45^\circ$ ,  $90^\circ$  and  $132^\circ$  for D0.94, D1.12, and D1.03 respectively. We note however that this choice of axis may be perturbed by extrinsic contributions to the polarization phase and hence our assignment of the crystal coordinates may not be accurate. Nonetheless, we can compare qualitatively the orthogonal coefficients  $L_1/L_2$  that we extract from the polarization dependence with the calculated conductivities.

To arrive at the experimental parametrization of the photocurrent introduced in the main text (which is measured along  $y'$  axis)

$$V_y' = L_1 \sin 2\theta + L_2 \cos 2\theta + D \quad (S9)$$

we project angular harmonics of Eq. 4 above to obtain

$$\begin{aligned} L_1 = & \frac{1}{4}(\sigma_{xxx} - \sigma_{xyy} + \sigma_{yxy} + \sigma_{yyx}) \cos \beta \\ & + \frac{1}{4}(-\sigma_{xxx} + \sigma_{xyy} + \sigma_{yxy} + \sigma_{yyx}) \cos 3\beta \\ & - \frac{1}{2}(\sigma_{yxx} - \sigma_{yyy} + (\sigma_{xxy} + \sigma_{xyx} + \sigma_{yxx} - \sigma_{yyy}) \cos 2\beta) \sin \beta \end{aligned} \quad (S10)$$

$$\begin{aligned} L_2 = & \frac{1}{4}(-\sigma_{xxy} - \sigma_{xyx} + \sigma_{yxx} - \sigma_{yyy}) \cos \beta \\ & + \frac{1}{4}(\sigma_{xxy} + \sigma_{xyx} + \sigma_{yxx} - \sigma_{yyy}) \cos 3\beta \\ & + \frac{1}{2}(\sigma_{yxy} + \sigma_{yyx} + (-\sigma_{xxx} + \sigma_{xyy} + \sigma_{yxy} + \sigma_{yyx}) \cos 2\beta) \sin \beta \end{aligned} \quad (S11)$$

$$D = \frac{1}{2}(\sigma_{yxx} + \sigma_{yyy}) \cos \beta - \frac{1}{2}(\sigma_{xxx} + \sigma_{xyy}) \sin \beta \quad (S12)$$

for the three coefficients of Eq. 9. In the above equations,  $\theta = \phi - \beta$  denotes the electric field polarization with respect to the experimental axis. We highlight here that in addition to the two polarization axis-dependent components, the general analysis of Eq. S8 gives a coefficient, here called  $D$ , that is, independent on the direction of polarized light.

### S6c - Symmetry dependence

In cases where the system lacks symmetry, there are six distinct components in the second-order conductivity tensor. However, in the absence of strain, the system exhibits  $C_{3z}$  symmetry, which imposes the following constraints on the components of the second-order conductivity.

$$\sigma_{xxy} = -\sigma_{yyy} = \sigma_{yxx} \quad (S13)$$

$$\sigma_{xxx} = -\sigma_{xyy} = -\sigma_{yxy} \quad (S14)$$

When  $C_{2y}$  symmetry is also present, only one independent component (i.e.,  $\sigma_{xxy}$ ) remains nonzero. To obtain nonzero  $\sigma_{xxx}$  and  $\sigma_{xyy}$  components, we introduced a layer-dependent sublattice offset,  $\Delta_1 \neq \Delta_2$ . The  $\sigma_{xxx}$  component exhibits peaks at two frequencies, which are close to the sublattice offset values  $\Delta_1$  and  $\Delta_2$ . In contrast, the  $\sigma_{xxy}$  behaviour is characterized by a single peak in frequency, coinciding with the average gap, as discussed in Ref. 7. Furthermore, the qualitative features of these plots are not influenced by a change in the sublattice offset  $\Delta_1$  and  $\Delta_2$ , which only shifts the peaks in frequency space, as shown in Fig. S9.1 and Fig. S9.2.

In the remainder of this supplement section, we will focus on the dependence of the individual conductivity coefficients as their behaviour can be more straightforwardly traced back to the microscopic origins. We note for reference the expression for the photocurrent and the corresponding experimental coefficients when the  $C_{3z}$  symmetry is present in the system, e.g., the system is unstrained:

$$L_1^{C_{3z}} = -\sigma_{xxx} \cos 3\beta + \sigma_{yyy} \sin 3\beta \quad (S15)$$

$$L_2^{C_{3z}} = -\sigma_{yyy} \cos 3\beta - \sigma_{xxx} \sin 3\beta \quad (S16)$$

$$D^{C_{3z}} = 0 \quad (S17)$$

Specifically, if  $C_{3z}$  symmetry is present, the polarization-independent term  $D$  vanishes and the two components,  $L_1$  and  $L_2$ , take also a more straightforward form where the relation to the non-linear conductivities of Eq. 10,11 is more explicit. We note however that in our experiments a finite  $D$  is always present and share some similarities with  $L_1/L_2$  we attribute to strain effects (S9).

#### S6d - Filling dependence

We now discuss the photocurrent behaviour as a function of filling. For simplicity of the analysis, we focus on the case of  $C_{3z}$  symmetry as we found that it qualitatively reproduces experimental trends and could be observed experimentally through polarization dependent measurements on D1.5 (see Supplementary Section 11). The two components  $\sigma_{xxx}$ ,  $\sigma_{xxy}$  exhibit remarkably distinct responses to changes in filling.

Specifically, the  $\sigma_{xxx}$  component reaches its maximum value at zero filling but gradually decreases with increasing non-zero fillings, as depicted in Fig. S9.1. This behavior can be attributed to a substantial contribution from the region where the gap forms due to a finite sublattice offset. Consequently, its magnitude diminishes when transitions from these regions are Pauli-blocked at non-zero fillings. Additionally, it is worth noting that the sign of the xxx component remains unaffected by the filling value and is solely determined by the sign of the difference between  $\Delta_1$  and  $\Delta_2$ .

Conversely, the  $\sigma_{xxy}$  component exhibits a markedly distinct response to variations in filling. Initially, it possesses a small, non-zero value at the charge neutrality point (CNP), which increases with higher filling, its sign depending on the filling value. This dependence on sign becomes noticeable only at considerably large fillings, approximately around,  $\nu \approx 2$ , as depicted in Fig. S9.2.

The two independent components of shift-current conductivity demonstrate a behavior similar to the  $L_1$  and  $L_2$  components of photocurrent discussed in the main text. However, it's important to note that the observed trend of sign variation in  $\sigma_{xxy}$  with filling is too weak to account for the sharp polarization phase slip observed in the photocurrent at the CNP, as shown in Fig. 3e of the main text.

A noteworthy characteristic of the  $\sigma_{xy}$  component is its peak frequency, which displays a subtle dependence on the filling values. As shown in Fig. S9.2, the peak position remains unchanged for filling values of opposite signs and equal magnitudes but starts to increase with greater magnitudes of filling values. Furthermore, unlike  $\sigma_{xx}$ , the sign of  $\sigma_{xy}$  remains unaltered when  $\Delta_1$  and  $\Delta_2$  are interchanged.

### S6e - The Role of interactions

Electron-electron interactions in twisted bilayer graphene can significantly affect the band structure and quantum geometry of electronic bands. Previous work in Ref. 7 has established that these interactions can significantly enhance the shift current conductivity compared to non-interacting band structures. In these calculations, we account for electron-electron interactions by incorporating Hartree corrections as we expect that at temperatures larger than  $\sim 10\text{K}$ , the role of exchange phenomena is suppressed. The effect of Hartree corrections is particularly pronounced in  $\sigma_{xy}$  at finite fillings, resulting in a sharp sign-changing behavior around the charge neutrality point (CNP), where the sign of  $\sigma_{xy}$  depends on the filling's sign. In contrast,  $\sigma_{xx}$  remains largely unaffected by Hartree corrections, primarily due to the main correction originating around the K/K' points shifting these points upwards in energy compared to the  $\Gamma$  point, but retaining the overall Dirac-like linear dispersion. As a result of this, effectively, the band structure near the  $\Gamma$  point becomes flattened (See Ref. 7 and references therein and the discussion in the main text) compared to the K/K' points. Consequently,  $\sigma_{xx}$  consistently retains a peak feature at the CNP.

These substantial modifications in  $\sigma_{xy}$  offer a plausible explanation for the observed polarization phase drift phenomena in the photocurrent experiment, as illustrated in the right panel of Fig. 3. These changes arise from the interaction-induced modifications to the shift current integrand, which exhibit opposing trends for opposite fillings, as indicated by the green circles in Fig. S9.3.

### S6f - The Role of strain

We also study the role of strain (Fig. S9.4) on different components of second-order conductivity. We apply opposite strain to two layers following the formalism in Ref. 17. Aside from the overall enhancement in the magnitudes of in conductivities, we notice several additional consequences. In particular, the peaked response characteristic to  $\sigma_{xx}$  is seemingly suppressed and attains a sign-changing behaviour similar to  $\sigma_{xy}$ . In fact, both conductivities have very similar qualitative behaviour with filling factor, which would lead to a more rigid polarization dependence – a photoresponse that depends weaker in the polarization direction of the incident electric field. Both these findings are consistent with our experimental observations regarding the measurements of D1.03 (Fig. 5 main text). In this sample, the peaked response at CNP is less prominent and the polarization dependence is significantly more rigid. Therefore, we attribute these marked differences to a larger inherent strain in D1.03. The difference with other magic-angle samples we believe lies in the presence of the supermoiré potential which acts as a magnifying glass<sup>18</sup> amplifying strain. Our theory is evidenced by the near-field photocurrent maps which show striped phases at length scales of several 100 nanometres. We note that whilst this qualitative analysis offers some explanation to the marked difference in the polarization dependence in D1.03, the second-order superlattice may be playing an even more pivotal role that has so far not been addressed on any theoretical footing.

## S7 – Quantum transport measurement of the $2\omega$ response in twisted bilayer graphene

Some of the most prominent intraband contributions to the second-order conductivity in twisted bilayer graphene has been shown to originate from quantum rectification effects, in-which the internal symmetry of the system causes a DC or  $2\omega$  response when excited with an AC signal of frequency  $\omega$ . These effects can be measured using  $2\omega$  frequency measurements. In our THz measurements, such effects can be observed in the DC response to THz excitation. In other words, the intraband THz response can be thought of as the non-linear quantum rectification in the high frequency limit and therefore should have a similar response to the  $2\omega$  quantum transport experiments in the lower frequency limit. To test this, we performed non-linear quantum transport experiments using the same probes from which we measure photoresponse (Fig. S10). The geometries are indicated in the schematics below the figure. The black data plotted on left axis correspond to the quantum transport experiment with current driven along the longitudinal direction of the Hall bar, whereas the red on the right axis plots the photoresponse measured with the same pair of contacts under photoexcitation with polarization direction aligned in the longitudinal direction of the Hall bar. We find the two responses are quite different. The only finite response of the  $2\omega$  response is located close to full filling, whilst in the THz photoresponse it is finite also in the bands. Furthermore, a small sign-changing-like feature can be seen in the  $2\omega$  response close to charge neutrality, whilst the photocurrent is peaked. This comparison shows the measured photoresponse is rather different to the  $2\omega$  response and originates from a distinct mechanism.

## S8 – Temperature dependent quantum transport measurements characterizing the gaps at full filling

In the main text, we claim that the terahertz excitations in our system lie within the flat bands and excitations between flat and remote bands are negligible because our excitation energies (3-10 meV) are smaller than the band gaps separating the flat and remote bands. To verify this, we present in Fig. S11 quantum transport measurements on the studied devices and extract the band gaps from temperature dependent measurements of the resistive states at full filling of the moiré Brillouin zone  $\nu = \pm 4$ . Figure S11a plots the conductivity ( $\sigma_{xx}$ ) as a function of temperature measured within the insulating states for filling factors  $\nu = \pm 4$ . We find that the conductivity increases dramatically approaching higher temperatures following an exponential trend expected for activation behaviour. The extracted gaps are plotted in Fig. S11b for the studied samples. In general, the gaps are larger than the maximum excitation wavelengths employed in the main text (10 meV). This is clear for all electron doping. For hole doping, in D0.94 and D1.03 the gaps are significantly suppressed. This behaviour may be attributed to the alignment to the underlying hBN substrate which can strongly perturb TBGs electronic spectrum. We note that the small gaps on the hole doping side may result in some contributions to the photoresponse from flat-remote band transitions when the system is doped close to  $\nu = \pm 4$ . Our experimental data, however, still suggests they may be negligible since the response generally attains e-h asymmetry close to full filling mimicking what is observed for electron doping when the gaps are large and flat-remote band are certainly suppressed. Moreover, given the finite band widths  $\sim (10 - 20 \text{ meV})$  expected for magic-angle TBG, those transitions can be safely ignored in our analysis of the shift current contributions close to charge neutrality (Fig. 3 of main text).

## S9 – Alignment angles of twisted bilayer graphene to hBN extracted from optical images

As explained in the main text, manifestation of the bulk photovoltaic effect requires the breaking of inversion symmetry. In twisted bilayer graphene, in-plane components of the conductivity are non-zero only in the case of breaking  $C_2T$  symmetry. This is typically achieved via alignment to an underlying hexagonal-boron nitride substrate which breaks the sub-lattice degeneracy of graphene imposing a

C3 symmetry on the twisted graphene system. Aside from our photocurrent experiments, signatures of this symmetry breaking are usually observed as gap openings in the flat-band due to the breaking of the sub-lattice symmetry<sup>19</sup> and additional satellite peaks in the resistivity originating from the graphene/hexagonal-boron nitride moiré superlattice<sup>20,21,22</sup>. This is clear in our D1.03 (Fig 1d of main text) whilst our other devices show no signatures in quantum transport measurements of alignment to the hBN substrate. Optical images (Fig. S12) highlight that our device may have alignment angles of approximately 4°, usually far from what is expected to have any significant influence on TBGs electronic spectra. However, recent work has demonstrated these alignment effects can manifest even for these large twist-angles due to complex lattice reconstruction effects leading to the breaking of C<sub>2</sub>T symmetry (See Supplementary Section 10).

### **S10 – Tight-binding calculations of our twisted bilayer graphene encapsulated with hexagonal-boron nitride devices**

Our shift current calculations outlined above are generic for twisted bilayer graphene with a sub-lattice off-set, and hence with a gapping of their flat-bands. The results are generic regardless of the underlying mechanism breaking TBG's sub-lattice and corresponding C<sub>2</sub>T symmetry. In the main text we postulated the possibility of hBN alignment being one possibility for the possible symmetry breaking in our devices. Below, we perform tight-binding calculations showing the effects of hBN alignment for relatively large mis-alignment angles supporting this claim.

Using the alignment angles extracted from the optical images we proceed to calculate the band structure of TBG on hBN using an atomistic approach<sup>23</sup>. We consider a tetralayer structure, where our approach first involves the construction of an unrotated structure, where the bonds of graphene and hBN are parallel. The centre of the supercell has an AAAA stacking, where a carbon site of graphene and nitrogen (N) sites of top and bottom hBN share the same in-plane position  $(x, y) = (0, 0)$ . Next, we rotate the top layer graphene ( $\theta_{\text{TBG}}$ ), the top and bottom layer hBN ( $\theta_{\text{TOP}}$ ,  $\theta_{\text{BOT}}$ ) with respect to the bottom layer graphene.  $\theta_{\text{hBN}}$  is the twist angle between hBN layers. We use positive angles to indicate counterclockwise rotations in our notation. It is important to note that, to keep the periodicity of these structures, the lattice constant of hBN is slightly modified. After constructing a commensurate supercell, we use semi-classical molecular dynamics, which is implemented in LAMMPS<sup>24</sup>, to fully (both in-plane and out-of-plane) relax the graphene layers. For intralayer and interlayer interactions between graphene, we use the adaptive intermolecular reactive empirical bond order potential (AIREBO-M)<sup>25</sup> and the interlayer potential (ILP)<sup>26</sup>, respectively. We use the same RDKC potential for the C-B and C-N with interlayer interaction of 60% and 200% with respect to the original C-C interaction, respectively<sup>27</sup>. The hBN layers are fixed in a flat configuration to mimic a few layers substrate. We assume that the relaxed structures keep the periodicity of the rigid cases. Once the TB Hamiltonian is constructed, we calculate the electronic properties of the encapsulated TBG/hBN system.

The two middle narrow bands of pristine TBG are degenerate at the K and K' points, which are protected by C<sub>2</sub>T symmetry. Here, C<sub>2</sub> represents the two-fold rotation operator and T represents the time-reversal operator. However, the presence of an hBN substrate induces a mass term (due to a sublattice polarization) that breaks C<sub>2</sub>T symmetry, resulting in the opening of a gap in the Dirac cones. This gap is well-known to exist in samples where the hBN is nearly aligned with its adjacent graphene layer. Previous theoretical studies have suggested that this gap persists even for angles between the hBN and graphene that are far from alignment<sup>27,28</sup>. The lattice structure of the tetralayer system and the degree of alignment for the two samples are shown in Fig. S13a. The corresponding band structures are depicted in Fig. S13b and Fig. S13c, where a breaking of inversion symmetry is evident even if each hBN layer is far from alignment. According to Refs <sup>27,28</sup> the persistence of the mass gap is

a combined effect of the large on-site potentials of the boron and nitrogen sites and the large strain fields induced by relaxation in the TBG lattice structure. We found that in both D1.12 and D0.94 samples, assuming the alignment angles extracted from optical images in Fig. S12, the encapsulation induces a small mass gap of about  $\sim 4\text{--}6$  meV which is strongly sensitive to the degree of alignment between hBN and its adjacent graphene layer. If the degree of alignment is reduced, this mass gap is enhanced.

The hBN generates an effective superlattice potential acting in the adjacent graphene layer. If graphene/hBN is commensurate, then the potential can be expanded in terms of the graphene reciprocal space basis<sup>29</sup>, resulting in an uniform and periodic position-dependent mass gap whose magnitude strongly depends on the degree of alignment between graphene and hBN. In addition, an interaction induced Hartree-Fock gap has been predicted at charge neutrality point<sup>30</sup>, by fitting a continuum model to the TB bands in Fig. S13 and by using the procedure described in Ref. <sup>19</sup>, we have found that the mass gap is increased by 2-4 meV for both twist angles depending on the value of the dielectric constant. If TBG is incommensurate with hBN, the superlattice potential becomes a non-periodic spatially varying field. For small degrees of alignment, the systems tend to stabilize the commensurate condition<sup>31</sup> and the periodic mass gaps are well defined. For large angles we expect local variations of the sublattice polarization. A similar effect is used to describe position-dependent band topology in TBG encapsulated with hBN<sup>32</sup>.

### **S11 – Second-order response in D 1.5 and measurement of C3 symmetry**

In second-order response, the underlying symmetry of the system can be mapped out by measuring its angular dependence, which is typically done using a circular mesa or the so-called sun-flower geometry, that allows measurement between different contact pairs for a fixed driving field. In our photocurrent measurements, we measure Hall bar devices which prevent a full mapping of the symmetry in this way. However, the symmetry may also be measured in Hall bar devices, by fixing the choice of contacts and rotating the polarization direction. In this case, the response always exhibits a  $180^\circ$  periodicity, but the polarization phase changes between the choice of contacts and should maintain a strict mapping reflecting the underlying symmetry of the system. Here we provide evidence for an intrinsic C3 symmetry in a non-magic angle device (D1.5) we attribute to the hBN substrate. Figure S14a plots the resistivity of the device as a function of gate voltage. In contrast to the magic-angle samples, no integer filling features can be observed. It does however exhibit marked insulating states from which we extract gaps of around 50 meV and a twist-angle of  $1.5^\circ$ . The sharp peak at CNP is not activated but rather weakens slowly with increasing temperature characteristic of a zero density of states rather than a band gap. The sample hence exhibits no signature of C3 symmetry breaking via quantum transport. Nonetheless, as in the magic-angle samples, it obtains a complex polarization-dependent response. Figure S14b plots the photoresponse measured between the pair of contacts indicated in the schematic inset referred to as “longitudinal geometry”. It shows strong peak-like and sign-changing structures with the polarization of light. The orthogonal direction shows similar response except it is phase shifted by  $45^\circ$  (Fig. S14,c,e). Analysing the coefficients L1 and L2 we find that this phase shift is exactly the value that would be observed in a C3 symmetric system and serves as a photocurrent fingerprint of broken inversion symmetry. This mapping between two orthogonal pairs extends for finite doping close to the CNP (Fig. S14d,f) reaffirming the symmetry relation. Upon close inspection, sharp polarization phase drifts in the vicinity of the CNP can also be observed (Fig. S14g) providing further evidence for its intrinsic second-order nature. We note however that the symmetry condition seemingly breaks down close to full filling and within the remote bands we attribute to extrinsic contributions to the polarization dependence (see Supplementary Section 5). This analysis demonstrates that multiple mechanisms may contribute to

the polarization phase and may be perturbing our interpretation of the principal/crystallographic axis. However, the ones that are intrinsic to TBGs quantum geometry may still be distinguished via changes in the polarization phase induced by doping. These measurements hence provides unambiguous evidence of the second-order response and intrinsic C3 symmetry of TBG on hBN. In our magic-angle samples we find no strict symmetry relation can be found between different contact pairs which we attribute to its fragile lattice structure, in which domains of varying alignment and even sub-lattice offsets can exist within a single sample and mixing with extrinsic contributions. Nonetheless, within a single choice of contacts, strong changes in their polarization phase induced by doping not only confirm its origin in the second-order response, but also reveal the quantum geometry induced by interaction effects.

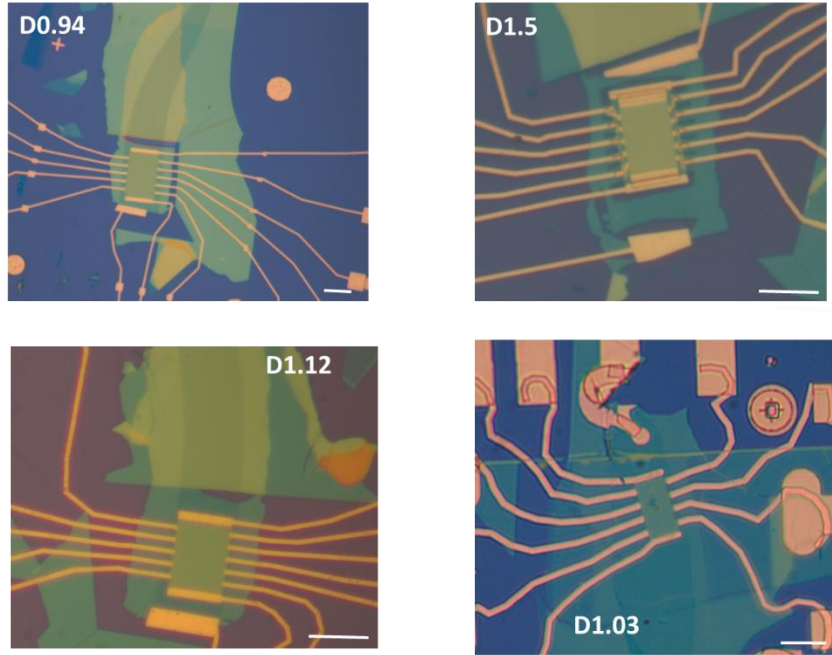
## **S12 – Measurements between different contact configurations in magic-angle samples**

As demonstrated in the Supplementary Section 11, measurements between different contact pairs provide further evidence for the bulk origin of the effect and allows one to discern the symmetry of the system. In our non-magic angle sample (D1.5), we found that the photoresponse measured between two orthogonal pairs of contacts respects that expected for a C3 symmetric system, in which a 45° phase shift is measured in the polarization phase when changing contact pairs by 90°. In our magic-angle samples, this symmetry relation was not so easy to demonstrate and the behaviour was rather more complex. Figure S15 plots the polarization phase measured between different contact configurations in D1.12; the corresponding measurement geometries are sketched on the right-hand side. G3 and G6 depict two measurement geometries where the current axis is orthogonal to each other as in the case of D1.5 (Supplementary Section 11). The two measurements however are rather different, in which the polarization phase of G6 varies much weaker than G3 and no connection can be made between their polarization dependences. This discrepancy suggests that the measurement G6 contains additional contributions to the polarization dependence that may be occurring locally, for example, near-field enhancements which may be larger because of the significantly larger contacts. With the rest of the working contacts available, we performed similar measurements (G3-G6) for the same wavelength and temperatures and notice several further distinct observations. In all contact configurations, marked changes in the polarization phase can be observed. In G1, G2 and G3 the overall trends in the polarization phase are similar. This demonstrates that the changes in the polarization phase are not unique to a certain choice of contacts and provides ample evidence for its bulk origin. Second, the polarization phase drifts at CNP are distinct to certain regions of the device. This suggests that the CNP response may have some contributions from interaction effects which is truly local, in agreement with our quantum transport measurements that cannot discern a gapped state. On the other hand, the presence of the phase drift at CNP may be unique to local unstrained regions of the device. Third, in one geometry the polarization phase drifts in the opposite direction to the others (G5). Physically, this symmetry condition requires one of the conductivities to change sign spatially. This may be the case in the presence of stacking faults which may reverse the sub-lattice offsets and the sign of  $\sigma_{xx}$  accordingly (see Supplementary Section 6). These complex behaviours suggest the symmetry of magic-angle TBG is far more complex than the simple C3 case as expected for rigid TBG on hBN. We postulate that strain may be responsible for breaking further symmetries, and, combined with local structural variations may be responsible for the complex contact dependencies observed in this system, whilst extrinsic contributions to the polarization phase may also be perturbing the symmetry analysis.

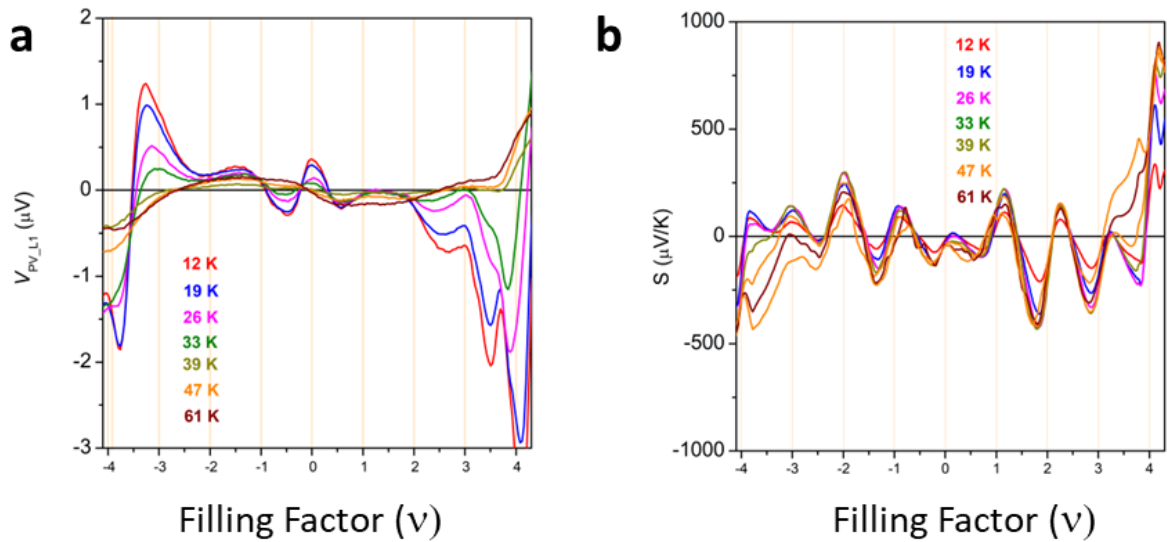
### S13 – Terahertz photodetection in magic-angle twisted bilayer graphene

As demonstrated in Supplementary Section 3, we observe that the THz response significantly increases around 0.7 THz, by nearly two orders of magnitude for certain doping levels. This is depicted more clearly in the inset of Fig. S16 which plots the wavelength dependence in  $D_{112}$  for  $\nu = -3.5$ . The sharp increase differing from a finite frequency Drude response<sup>33</sup> distinguishes its origin in the bulk photovoltaic effect and suggests we approach the interband resonances (inset of Fig. 1a). Moreover, the large response is consistently observed for different polarization directions (see Supplementary Section 2), suggesting that the enhancement is intrinsic to TBG rather than attributable to antenna effects. We find extrinsic voltage responsivities around 200 mV/W for optimal doping, which is large considering the device area is 1000 times smaller than the illumination area and a significant portion of the incident power is not absorbed. To estimate the capabilities of our TBG devices for terahertz detection, we project the possible noise equivalent powers (NEP) that may be obtainable in optimized devices with antenna integration (Fig. S16). For this, we estimate the total power incident on the sample by calculating the fraction of the incident terahertz beam that is occupied by the device. Considering the wavelength at 0.7 THz = 400  $\mu\text{m}$ , a numerical aperture (NA) of 0.12, the diffraction limited beam spot diameter is estimated to be 1.8 mm. Given our devices are at most 10 x 30  $\mu\text{m}$  in size, we find the device area ( $3\text{E-}10\text{ m}^2$ ) around 8000 times smaller than the area of the focused gaussian beam ( $2.5\text{E-}6\text{ m}^2$ ). Hence, the power axis on Fig. S16 is normalized by this factor 8000 to estimate the potential responsivities and NEPs obtainable for example in a device with integrated antennas or if the device active area was up-scaled. Because the device operates under zero-bias voltage, we attribute the intrinsic noise floor to Johnson noise, with  $R = 10,000\ \Omega$  (two-probe resistance of our devices),  $T = 10\text{ K}$  and  $\Delta f = 1\text{ Hz}$ . Extrapolating from Fig. S16 we find intrinsic NEP =  $7\text{E-}13\text{ W/Hz}^{0.5}$ . The power dependent measurements in Fig. S16 was performed for  $\theta_p = 50^\circ$  illumination.

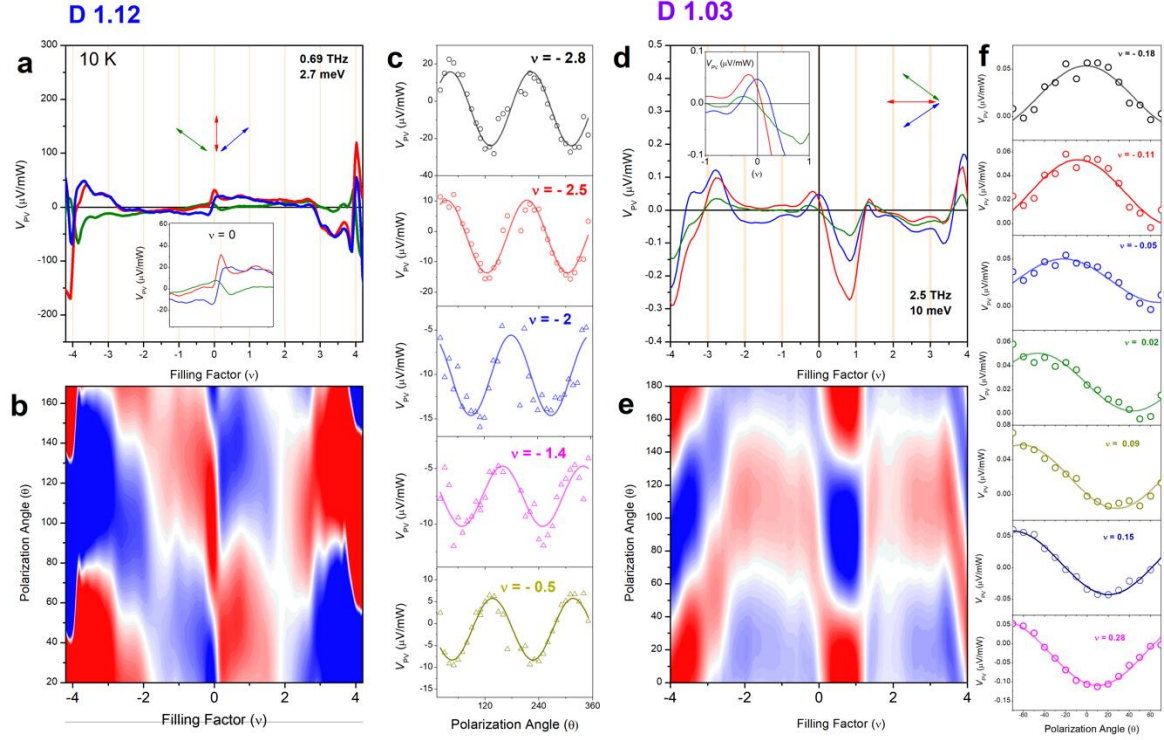
## Supplementary Figures



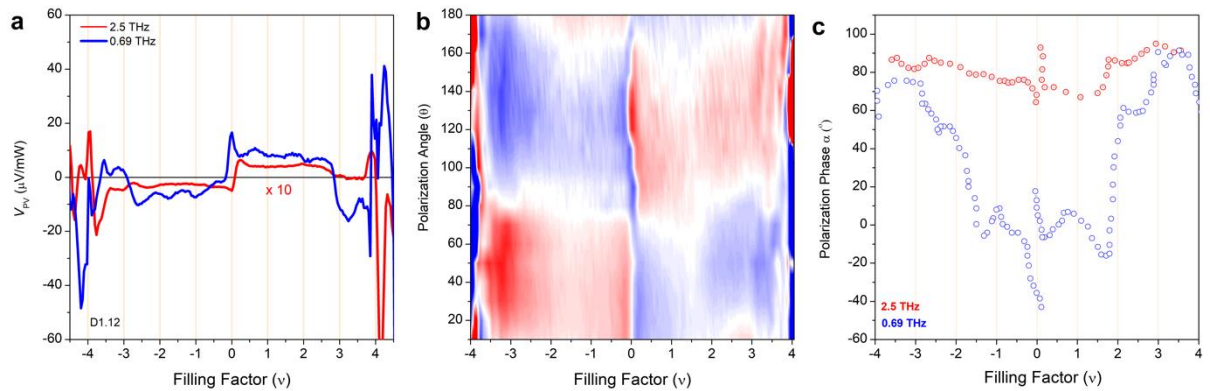
**Figure S1 Optical Images of studied devices.** The four panels show optical images of the studied twisted bilayer graphene devices in this work taken at 100 x or 50 x zoom. Scale bar corresponds to 10  $\mu\text{m}$ .



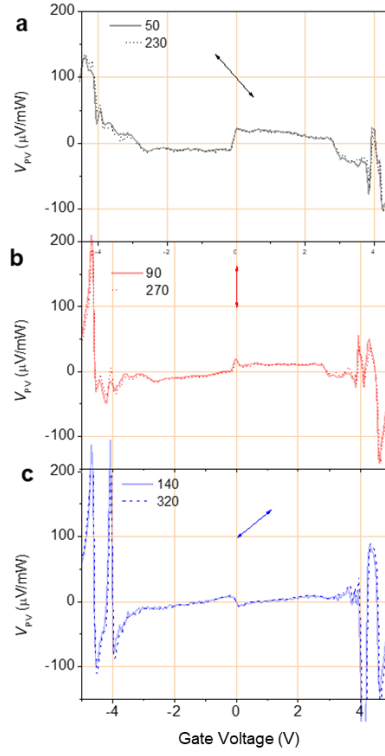
**Figure S2. a,** temperature-dependent measurements of the photovoltage  $V_{Lin}$  ( $\mu\text{V}$ ) as a function of filling factor ( $\nu$ ) measured in D0.94. **b,** Seebeck coefficient ( $S$ ) calculated via the Mott Formula (S1) plotted as a function of filling factor ( $\nu$ ) for the same temperature range as **a**.



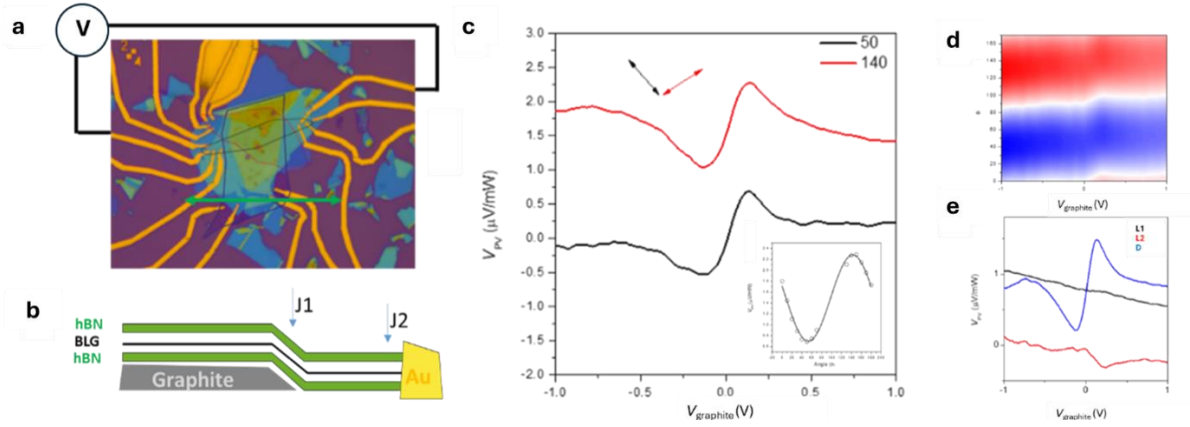
**Figure S3, data for D1.12 (a-c) and D1.03 (d-f).** **a**, photovoltage ( $V_{PV}$ ) as a function of filling factor ( $\nu$ ) measured in D1.12 for three different polarization directions ( $\theta_p$ ) at 0.7 THz (2.7 meV) excitation. Inset plots zoom around CNP. **b**, polarization dependent component of the photoresponse ( $V_{Lin}$ ) as a function of  $\nu$  and  $\theta_p$  for the same pair of contacts as in **a**. **c**,  $V_{PV}$  ( $\theta_p$ ) for different doping levels. Open circles are experimental data and solid lines are sinusoidal fits. **d**, same as **a** for D1.03. Polarization directions are chosen to highlight the emergence of the peaked response at CNP. **e**, same as **b** for D1.03. **f**, same as **c** plotted for doping levels in the vicinity of the CNP highlighting the phase drifting behaviour.



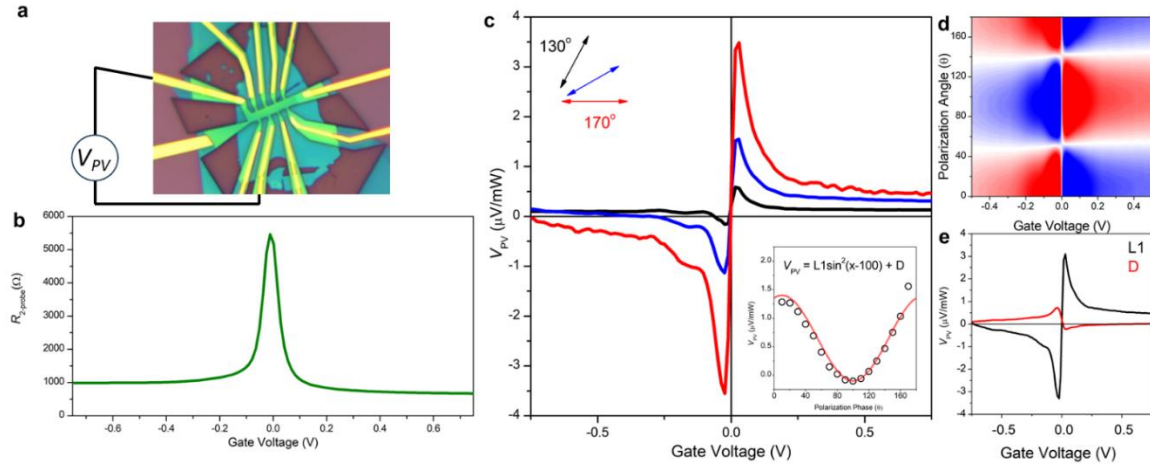
**Figure S4. Wavelength dependence of THz response.** **a**, Photovoltage ( $V_{PV}$ ) normalized by the incident power for a fixed polarization direction ( $\theta_p = 90^\circ$ ) measured at 0.69 THz (blue) and 2.5 THz (red); 2.5 THz data is multiplied by 10 to compare its functional form with 0.69 THz. **b**, Polarization dependent component of the photovoltage ( $V_{Lin}$ ) as a function of polarization angle ( $\theta_p$ ) and filling factor ( $\nu$ ) measured at 2.5 THz. **c**, Polarization phase extracted from maps Fig. S3b and **b** for 2.5 and 0.69 THz respectively.



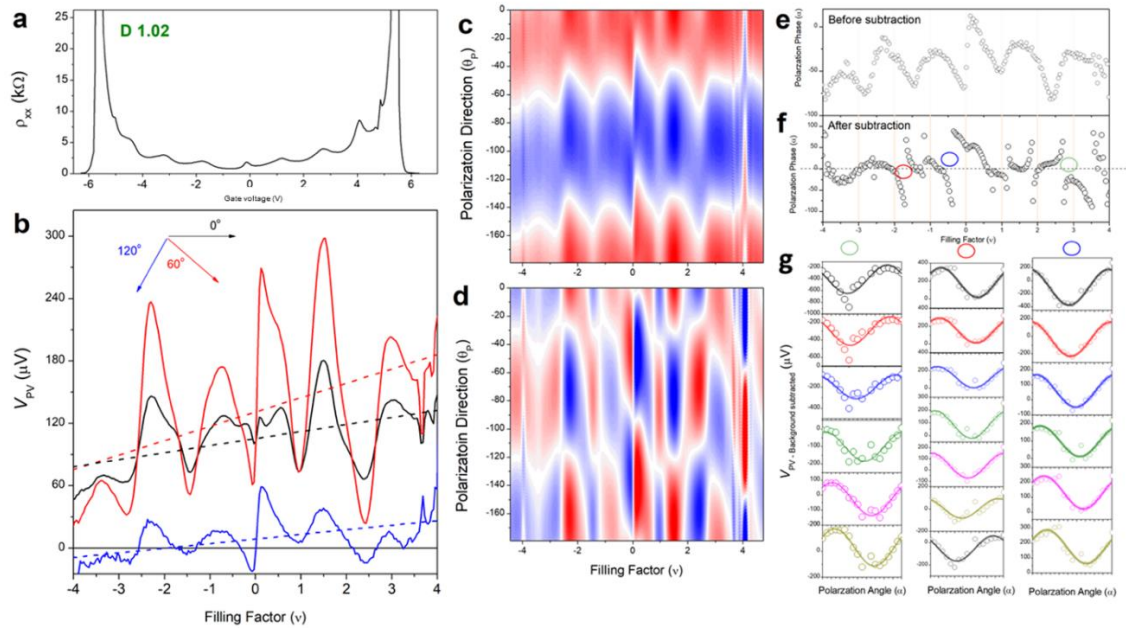
**Figure S5 Checking for artefacts.** (a-c) Photovoltage  $V_{PV}$  as a function of gate voltage for fixed polarization directions. Solid and dashed lines plot measurements made for polarization directions  $180^\circ$  apart.



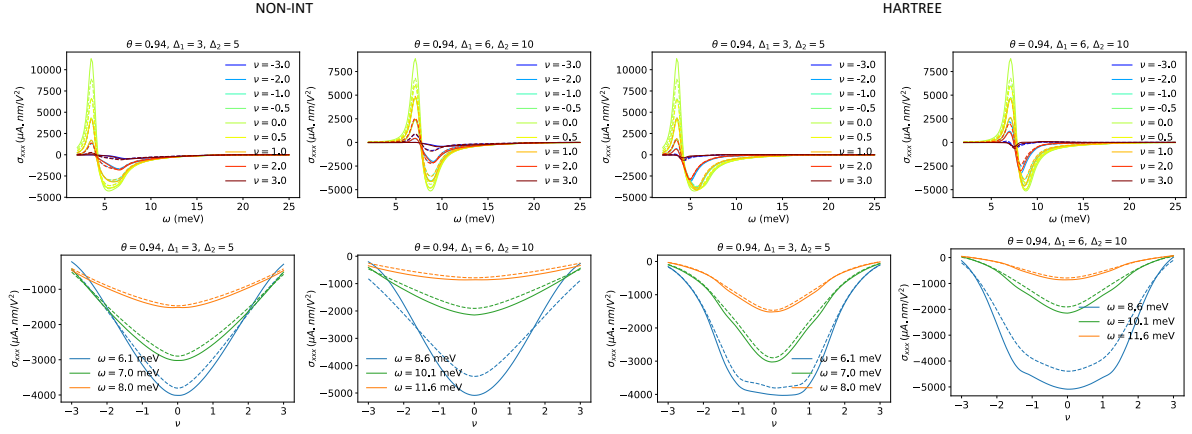
**Figure S6 Control measurements in Bilayer graphene.** **a**, Optical image of the studied device. Coloured triangles highlight the edges of bilayer graphene and top/bottom graphite gates. **b**, Schematic depicts the heterostructure and the presence of extrinsic junctions induced by gated/un-gated regions of the channel (J1) and local doping by gold contacts (J2). **c**, Photovoltage  $V_{PV}$  measured as a function of the bottom gate voltage  $V_{\text{Graphite}}$  plotted for two orthogonal (black and red) polarization directions. Inset: Polarization dependence plotted for a fixed gate voltage  $V_{\text{Graphite}} = 0.2$  V. Open circles are experimental data and solid line is a sinusoidal fit. **d**, polarization dependent component of the photoresponse plotted as a function of polarization direction ( $\theta_p$ ) and  $V_{\text{Graphite}}$ . **e**, Coefficients obtained from the fits required to describe the entire polarization dependent response. Black, red and blue plot L1, L2 and the offset D respectively.



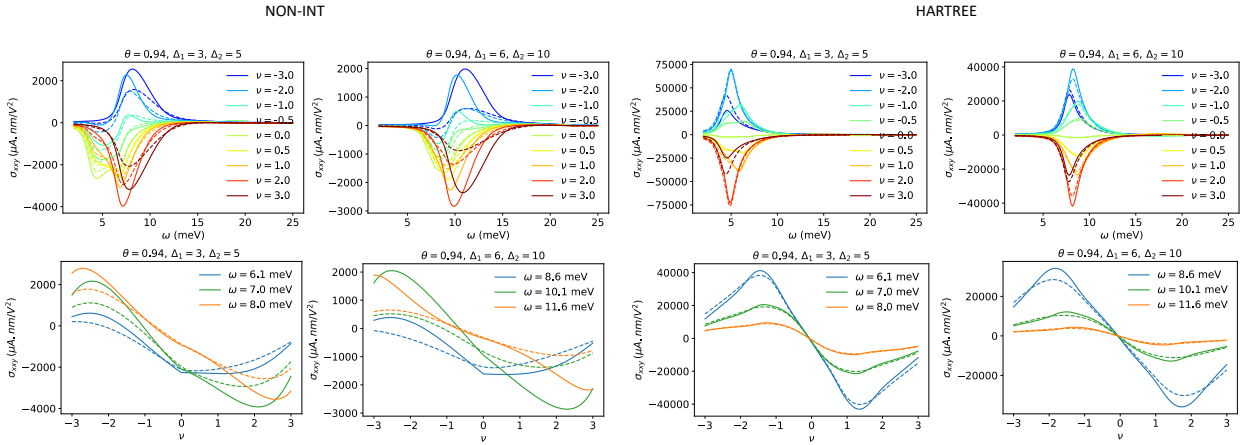
**Figure S7 Control measurements in single layer graphene.** **a**, Optical image of the studied device with the voltage probes used for measurements fixed. **b**, two-probe resistance ( $R_{2\text{-probe}}$ ) as a function of gate voltage measured between the same pair of contacts depicted in **a**. **c**, Photovoltage ( $V_{PV}$ ) measured as a function of gate voltage for three different polarization directions (depicted by coloured arrows). Inset: Full polarization dependence for a fixed gate voltage  $\sim 0.1$  V. Open circles plot experimental data and solid red line plots a sinusoidal fit, specifically a  $\sin^2$  function. **d**, polarization dependent component of the photoresponse  $V_{Lin}$  plotted as a function of the polarization direction ( $\theta_p$ ) and gate voltage ( $V_{Graphite}$ ). **e**, Extracted coefficients from the fits to the  $\sin^2$  function, L1 and the offset D. All measurements are performed at 10 K.



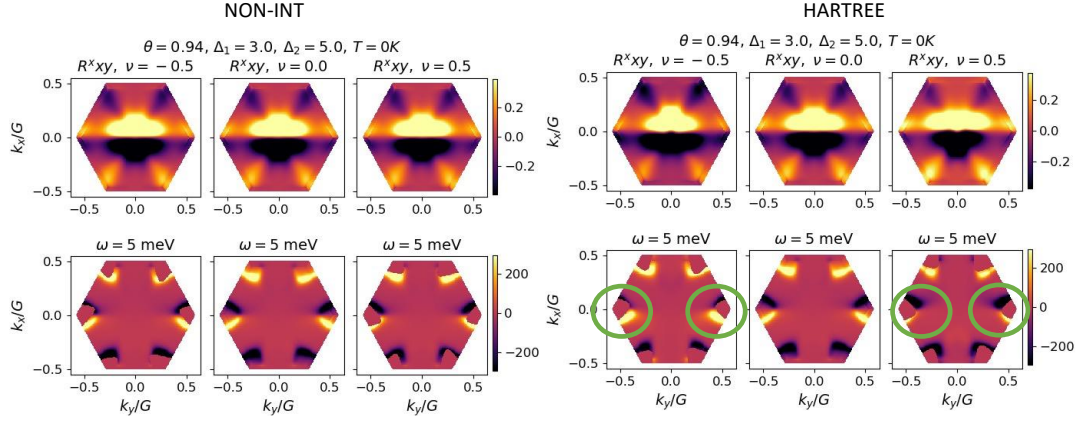
**Figure S8 Measurements in D1.02.** **a**, Resistivity ( $\rho_{xx}$ ) as a function of gate voltage measured in D1.02 at 10 K. **b**, photovoltage ( $V_{PV}$ ) as a function of filling factor ( $\nu$ ) plotted for three polarization directions. The solid lines are experimental data and the dashed lines are linear fits which define the background. **c**, Polarization-dependent component of the photoresponse ( $V_{Lin}$ ) plotted as a function of polarization direction ( $\theta_p$ ) and filling factor ( $\nu$ ). **d**, same as in **c** but after subtracting the linear backgrounds. **e, f** Polarization phases extracted before (**e**) and after (**f**) background subtraction. **g**,  $\Delta V_{PV}(\theta_p)$  plotted for different filling factor illustrating the phase drifting behaviours that can be discerned after background subtraction. The three columns illustrate the doping ranges highlighted by coloured circles on **f**.



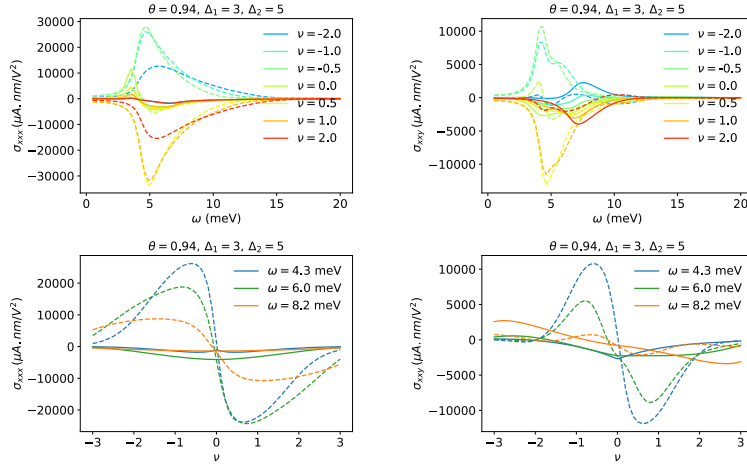
**Figure S9.1:** The  $\sigma_{xx}$  component of shift-current conductivity as a function of frequency (upper panel) and filling (lower panel) for TBG at twist angle  $\theta = 0.94^\circ$ . The two left columns depict the non-interacting case for different sets of layer-dependent sublattice offset ( $\Delta_1, \Delta_2$ ) at T=5 K (solid lines) and T=20 K (dashed lines). On the right side, we added the Hartree contributions from electron-electron interactions. We notice that Hartree corrections do not impact the  $\sigma_{xx}$  component significantly.



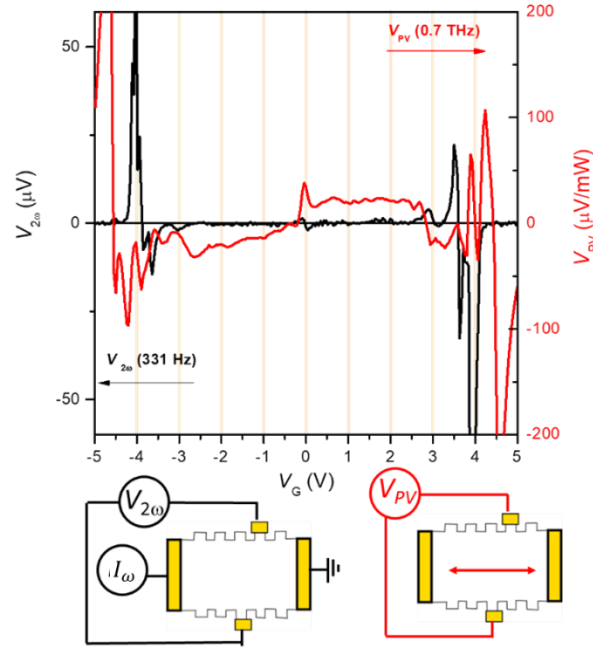
**Figure S9.2:** The  $\sigma_{xy}$  component of shift-current conductivity as a function of frequency (upper panel) and filling (lower panel) for TBG at twist angle  $\theta = 0.94^\circ$ . The two left columns depict the non-interacting case for different sets of layer-dependent sublattice offset ( $\Delta_1, \Delta_2$ ) at T=5 K (solid lines) and T=20 K (dashed lines). On the right side, we added the Hartree contributions from electron-electron interactions. We notice that Hartree corrections impact the  $\sigma_{xy}$  component significantly.



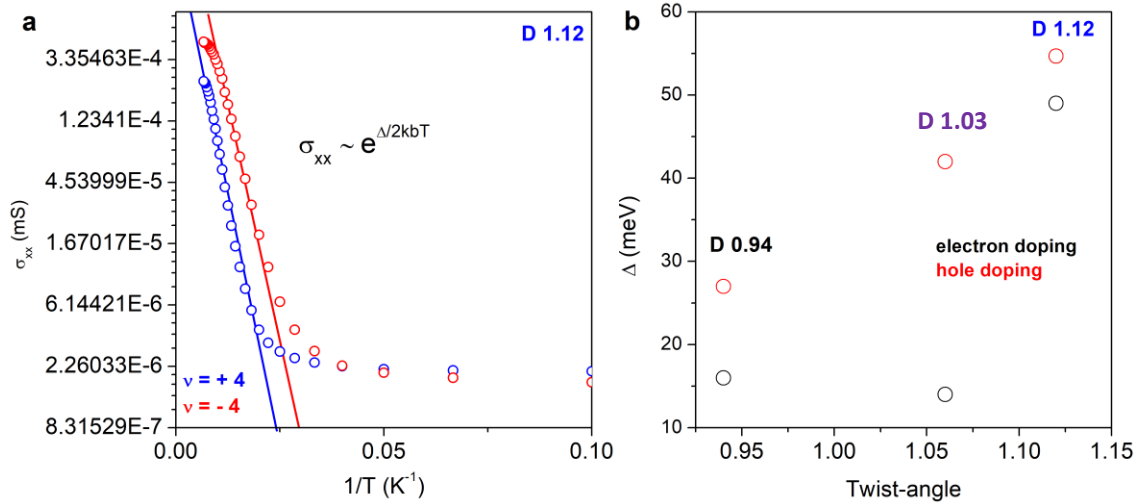
**Figure S9.3:** Shift-vector integrand for  $\sigma_{xxy}$  component for the non-interacting case (left panels) and with electron-electron interactions (right panels). The upper rows show the full integrand without the joint density of states at three different fillings of -0.5, 0, and 0.5. The lower rows include the joint density of states at frequency,  $\omega = 5 \text{ meV}$ . We notice that the  $\sigma_{xxy}$  component has almost equal regions of positive and negative values of shift-current integrand at zero filling. For the non-interacting case, a small but opposite imbalance in these two regions can be seen at finite filling of  $\nu = \pm 0.5$ . Interaction-induced effects significantly enhance this imbalance, as shown by expanding yellow and blue regions in green circles.



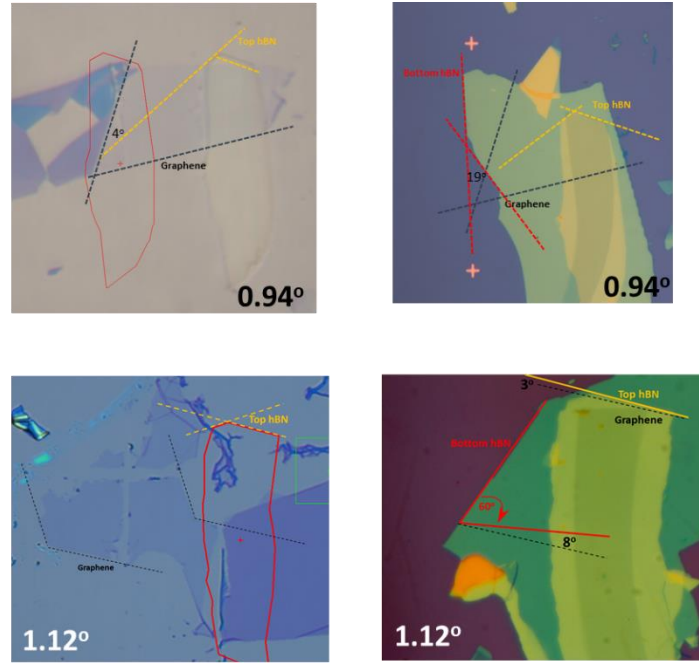
**Figure S9.4:** The  $\sigma_{xxx}$  and  $\sigma_{xxy}$  component of shift-current conductivity as a function of frequency (upper panel) and filling (lower panel) for TBG at twist angle  $\theta = 0.94^\circ$  with strain  $\epsilon = 0.1\%$  (dashed line) and without strain (solid line) at  $T = 5 \text{ K}$ . We notice that strain significantly enhances both the components. In particular, the  $\sigma_{xxx}$  component becomes very large and changes sign with the sign change in filling values.



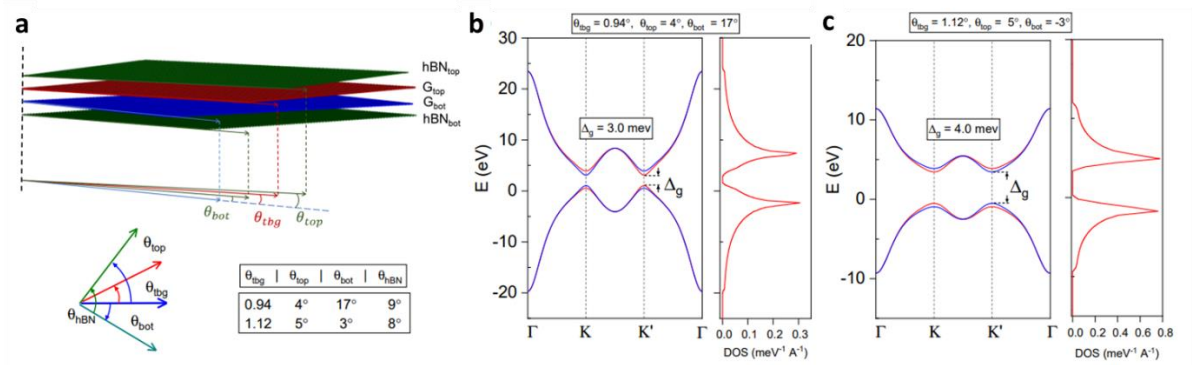
**Figure. S10 Measurement of second-harmonic response via AC quantum transport measurements.** Left axis: Voltage response ( $V_{2\omega}$ ) measured at twice the frequency of AC excitation ( $I_{\omega} = 331$  Hz) for different doping levels ( $V_G$  = gate voltage). Right axis (red): Photovoltage  $V_{PV}$  measured at 0.7 THz excitation with the polarization direction fixed in the same direction as the current drive measured in quantum transport (left axis). The schematics for both measurements are illustrated in black and red for quantum transport and photovoltage measurements respectively.



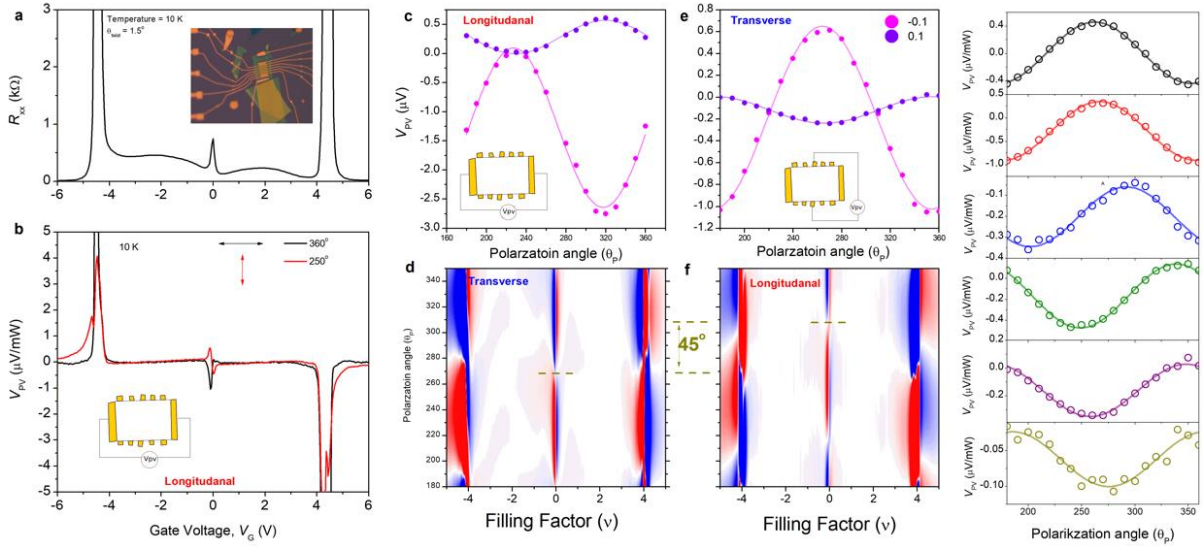
**Figure S11. Band gap extraction via quantum transport. a**, Conductivity ( $\sigma_{xx}$ ) plotted as a function of inverse temperature  $T^{-1}$  for fixed doping in the insulating states for electron (blue) and hole (red) doping in one of the studied devices. The open circles are experimental data, and the solid lines are fits to the exponential dependence from which the gaps ( $\Delta$ ) can be extracted. **b**, The band gaps separating flat and remote bands are extracted experimentally in the three studied samples of the main text, with the red and black circles plotting the gaps for hole and electron doping respectively.



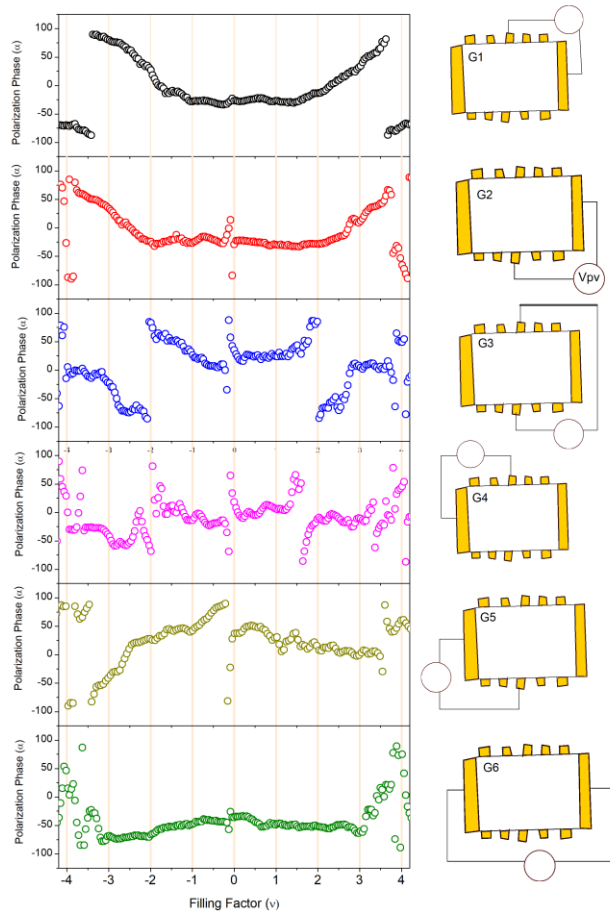
**Figure. S12 Angle alignment between straight edges of graphene and hexagonal boron nitride in D0.94 and D1.12 magic-angle devices.** Optical images of the heterostructures used in this study during the assembly process. The yellow, red and black lines traces top hexagonal-boron nitride (hBN), bottom hBN and graphene flakes respectively.



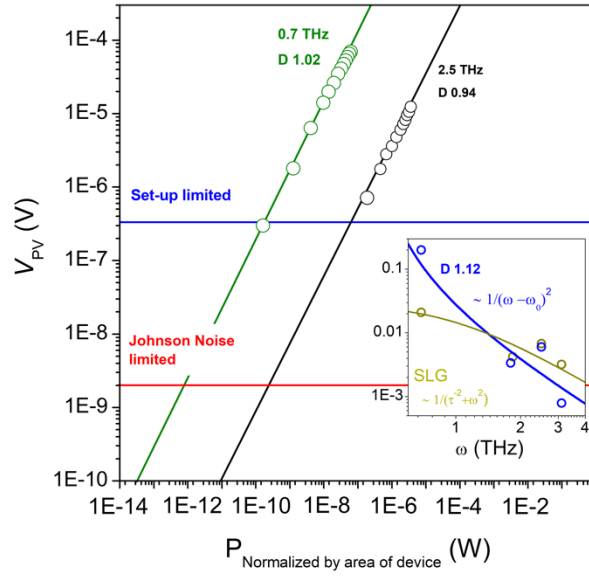
**Figure S13 tight-binding calculations of TBG on hBN.** **a**, schematic of the modelled TBG heterostructure. The Table indicates the measured alignment angles used for the calculations which are equal to or comparable to those extracted from optical images (Fig. S11). **b**, **c** calculated band structures for the heterostructures zoomed around the lowest lying bands. Pronounced gaps can be found ( $\Delta$ ) separating them. Right panels in **b** and **c** plot the density of states (DOS).



**Figure S14 Photocurrent measurements in D1.5 and proof of C3 symmetry.** **a**, resistivity ( $R_{xx}$ ) measured as a function of gate voltage ( $V_G$ ) in D1.05 at 10 K. Inset: optical image of the device. **b**, photovoltage ( $V_{PV}$ ) measured at 2.5 THz (10 meV) excitation for two orthogonal polarization directions indicated by black and red arrows. Inset plots a schematic of the device mesa and measurement geometry. **c,e**,  $V_{PV}$  measured as a function of polarization angle ( $\theta_p$ ) in two different geometries for the same doping levels. **d,f**, polarization-dependent component of the photoresponse  $V_{Lin}$  plotted for the same two geometries as a function of  $\theta_p$  and filling factor ( $\nu$ ). The yellow dashed lines indicate the polarization phase in the vicinity of the CNP. **g**, polarization dependence of the photovoltage ( $V_{PV}$ ) measured in the vicinity of the CNP. The open circles are experimental data and the solid lines sinusoidal fits.



**Supplementary Figure S15 Measurements between different contact pairs in D1.12** Polarization phase ( $\alpha$ ) measured as a function of filling factor ( $\nu$ ) between different contact pairs in D1.12 at a frequency of 0.7 THz (2.7 meV). Plots on the left are the extracted polarization phase for each of the different measurement geometries (G1-6) sketched on the right-hand side.



**Figure. S16 THz photodetection with magic-angle twisted bilayer graphene.** Photovoltage ( $V_{PV}$ ) as a function of power normalized by the device size relative to the diffraction-limited spot size ( $P_{\text{normalized}}$  by area of device) for fixed filling factor  $\nu = -0.5$  in D1.02 (green) at 0.7 THz and D0.94 at 2.5 THz. Open circles are experimental data and solid lines are linear fits. The dark blue and dark red lines represent the set-up limited and intrinsic device limited (thermal) noise-floors from which the estimated NEPs are extracted (see methods). **Inset:** wavelength dependence of the photovoltage measured in D1.12 at 0.7 THz for optimal doping  $\nu = -3.5$  (blue open circles) and fixed  $\theta_p = 50^\circ$ . The blue solid line plots function  $V_{PV} = a/(\omega - \omega_0)^2$  (solid red line) where  $a = 1 \times 10^{-5}$  and  $\omega_0 = 0.4$  THz. The green circles plot data measured in a SLG device for optimal doping and the solid green line plots a fit to a Drude-like response<sup>33</sup>,  $V_{PV} = a/(\tau^2 + \omega^2)$  dependence with  $a = 2.83 \times 10^{-5}$  and  $\tau = 1.02 \times 10^{-12}$  s.

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